Group 45.

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Question-1

Consider a car retail outlet that is giving away 3 cars (A, B, C) to winners of a contest. There are three winners (1st, 2nd, 3rd) and each gets only one car. Suppose that car A is the most valuable car for all the three winners. Car B, C are the second highest valued cars for winners 2 and 3 respectively. Winner 1 values cars B and C equally. How can the cars be assigned efficiently to the three winners? Provide the formal model and solve for the efficient assignment. Write down all the assumptions and show all the steps.

Answer-1

We are given:

Three winners {Agents} ; $N = \{1, 2, 3\}$

Three cars {objects} ; $M = \{A, B, C\}$

Each winner must be assigned exactly one car, and each car must be assigned to exactly one winner.

→ All winners rank Car A as the most-valuable.

→ Winner 2 ⇒ $B \succ C$

→ Winner 3 ⇒ $C \succ B$

→ Winner 1 ⇒ indifferent between B & C $(B \sim C)$

* My Assumptions

① Each winner receives exactly one car. [ONE to ONE]

② No car is assigned to more than one winner.

③ Preferences are COMPLETE and TRANSITIVE

④ Preferences are represented numerically for analysis.

⑤ No monetary transfers or side payments are allowed.

* Our Objective

To find a Pareto Efficient Allocation.

* Formal Model

→ Assignment Problem.

An allocation is a bijection :-

$f : N \to M$ such that each winner is assigned a unique car.

→ An allocation is pareto efficient if there is no other allocation that makes at least one winner strictly better off without making any other winner worse off.

* Analysis of All Possible Allocations

Since, all winners prefer A the most, we consider which winner receives A.

① CASE-1 : Winner 1 gets A.
- Winner 2 prefers B over C.
- Winner 3 prefers C over B.

Thus, we assign :
Winner 1 → A, Winner 2 → B, Winner 3 → C.

② CASE-2 : Winner 2 gets A.
- Winner 3 prefers C over B.
- Winner 1 is indifferent between B and C.

Thus, we assign :
Winner 1 → B, Winner 2 → A, Winner 3 → C.

③ CASE-3 : Winner 3 gets A.
- Winner 2 prefers B over C.
- Winner 1 is indifferent between B and C.

Thus, we assign :
Winner 1 → C, Winner 2 → B, Winner 3 → A.

* Check Envy for each Allocation

① ALLOCATION-1 : (1 → A, 2 → B, 3 → C)

Winner-1 : A → no envy (Winner-1 prefers A most)
Winner-2 : B → envies Winner-1
Winner-3 : C → envies Winner-1

→ Allocation-1 is not envy free.

② ALLOCATION-2 : (1→B, 2→A, 3→C)

Winner 1 : B → envies Winner-2
Winner 2 : A → envies no-one
Winner 3 : C → envies Winner-2

→ Allocation-2 is not envy free.

③ ALLOCATION-3 : (1→C, 2→B, 3→A)

Winner 1 : C → envies Winner-3
Winner 2 : B → envies Winner-3
Winner 3 : A → no envy

→ Allocation-3 is not envy free.

Therefore, no allocation is envy free because :
- → Car A is strictly best for all winners.
- → Only one can get car A.
- → So, the other two will always envy that winner.

But,
the structure of this problem suggests that the agents are ranked (1st, 2nd, 3rd) which introduces a notion of priority. So, Sometimes, we consider a refined fairness concept "Absence of Justified envy" in which "An agent's envy is considered justified only if they have higher priority than the person they envy.

Under Allocation-1 : (1→A, 2→B, 3→C)
Winner-2 envies Winner-1 but since, Winner-1 has higher rank, this envy is not justified.
Similarly, envy of Winner-3 is also not justified.
Under Allocation-2 and Allocation-3 ⇒
The envies are justified since, the winner to they envy are of lower rank / priority.

* SUMMARY OF ALL POSSIBLE ALLOCATIONS

Assignment (W1, W2, W3)	Pareto Efficient	Justified Envy?
1. (A, B, C)	YES	NO
2. (A, C, B)	NO {2 & 3 should trade}	YES
3. (B, A, C)	YES	YES
4. (B, C, A)	NO {1 & 2 should trade}	YES
5. (C, A, B)	NO {1 & 3 should trade}	YES
6. (C, B, A)	YES	YES

* CONCLUSION

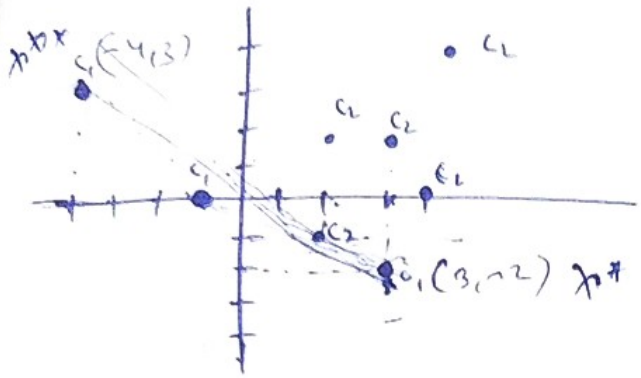
The allocation (1 → A, 2 → B, 3 → C) is the most intuitive as it is both pareto-efficient and justified envy free!

(Solution 2:)

To find the separating hyperplanes and analyze its Robustness to errors, we first organize the data into two classes:-

class (C_1): $\{(-4,3), (-1,0), (3,-2)\}$

class (C_2): $\{(3,2), (4,0), (2,-1), (4,4), (2,1)\}$



A separating hyperplane is defined by the equation $w_1x_1 + w_2x_2 + b = 0$. For a clean separation, we need all points in C_1 to yield a value > 0 and C_2 to yield a value < 0 (or vice versa).

~~Slope of~~

slope of $(x^* - x^{**}) = \frac{3-(-2)}{-4-3} = \frac{5}{-7}$ //

∴ Lets check for $5x_1 + 7x_2$

C_1 values $\{-20+21 = 1, -5+0 = -5, 15-14 = 1\}$. min = 1.

C_2 values $\{15+14 = 29, 20+0 = 20, 10-7 = 3, 20+28 = 48, 10+14 = 24\}$

min = 3. //

Since, the minimum value of C_1 (1) is less than the minimum value of C_2 (3).

hence, we can say that any value between 1 and 3 can serve as a boundry.

Separating hyperplane:

$5x_1 + 7x_2 = 2$ ✓ (choosing the midpoint).

Are there more than one hyperplane?

Yes. Because the two sets are linearly independent and finite, there are infinitely many hyperplanes.

We can slightly change the $(w_1$ & $w_2)$ or change the value of b: $(w_1x_1 + w_2x_2 = b)$ and still maintain the convex hull of the two sets.

hyperplane for error ≤ 0.2

To handle the error, a hyperplane must satisfy the condition that even if a point moves by ε in any direction, it stays on correct side.

Mathematically, the function margin must be greater than maximum possible shift:

$$\text{margin} > \varepsilon(|w_1| + |w_2|).$$

for our such hyperplane $5x_1 + 7x_2 - 2 = 0$

- The ~~for c~~ function margin (distance from the nearest points 1 or 3 to the boundary is 1.
- The Required Tolerance for $\varepsilon = 0.2$ is $0.2 * (|5| + |7|) = 0.2 * 12 = 2.4$

Since $1 < 2.4$, this specific hyperplane fails.

~~To find~~

Even, with the SVM, we check the closest point:
$(3, -2) \in C_1$ and $(2, -1) \in C_2$.

- In L_∞ distance (square error), these points are separated by 1 unit in x_1 and 1 unit in x_2.

- if error ≤ 0.2, the point $(3,-2)$ could be $(2.8,-1.8)$ and $(2,-1)$ could be $(2.2,-1.2)$.

- These error boxes are still separated. A Robust hyperplane exist such as $x_1 - x_2 = 4$, but it must also account for points like $(-1,0)$.

conclusion:-

for $\varepsilon \geq 0.2$, use a hyperplane that balances the gap, such as $x_1 + 2x_2 = 0$. However upon rigorous checking, the 'bottleneck' points suggests that while a separator exists for boxes, a single hyperplane that satisfies the error condition for all points simultaneously is highly constrained.

$\Rightarrow$ About error ≤ 0.5

if the error is ≤ 0.5

Point $P_4(3,-2)$ could actually be $(2.5,-1.5)$

and $P_5(2,-1)$ could actually be $(2.5,-1.5)$.

Because these two points from different classes can converge to the exact same coordinate $(2.5,-1.5)$ within the allowed error range, the dataset is linearly inseparable.

hence, No separating hyperplane can work for $\boxed{\varepsilon \leq 0.5}$

Question 3.

→ Let the divisible good be represented by $[0,1]$. There are 3 agents $i \in \{1,2,3\}$ with valuation functions

$$v_i : [0,1] \to \mathbb{R}_{\geq 0}$$

the functions satisfy the following conditions:

1] Non-negativity : $v_i(X) \geq 0$ for all $X \subseteq [0,1]$

2] Additivity : if $X \cap Y = \phi$, $v_i(X \cup Y) = V_i(X) + V_i(Y)$.

3] Continuity : Single points have zero value.

4] Normalization : $v_i([0,1]) = 1$

An allocation is a partition (A_1, A_2, A_3) such that

$$A_i \cap A_j = \phi \ (i \neq j), \quad \bigcup_{i=1}^{3} A_i = [0,1]$$

* It is envy-free if

$$v_i(A_i) \geq v_i(A_j) \quad \forall i,j \in \{1,2,3\}$$

★ Existence of ~~Envy~~ envy free allocation.

Let, $F = \{ u \in \mathbb{R}^3 : u_1 + u_2 + u_3 = 1, \ u_i \geq 0 \}$ be a simplex of all possible divisions.

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We will now define a function which maps each division to the preferred piece of agent. Using a labelling of triangulation of F, we can apply Sperner's Lemma which guarantees the existence of a fully labelled ~~sim~~ simplex. This corresponds to a division where each agent prefers a different piece.

Equivalently, by continuity of preferences we construct a continuous map from F to itself and obtain a fixed point corresponding to an envy free allocation by ~~on~~ applying ~~Brouwer~~ Brouwer fixed Point theorem. Hence an envy-free allocation exists for 3 agents.

★ Allocation procedure (self-ridge banuxay)

1] Agent 1 cuts the cake into three pieces A_1, A_2, A_3 such that $v_1(A_k) = \frac{1}{3}$ for all k.

2] Agent 2 trims the largest piece on basis of v_2, so that it will become the second largest piece. the cake is now split into cake 1 (the three pieces, one of which is trimmed) and cake 2 (trimmings).

3] Agent 3 selects his most preferred piece.

4] Agent 2 must select the trimmed piece, if 3 did not select it. If agent 3 already took the trimmed piece, 2 picks their preferred from what remains.

5] Agent 1 selects the last remaining piece. Since agent 1 originally cut the three pieces to be equal in their own eyes, no envy arises regardless of which piece they receive.

Dividing cake 2:

1. Let T be the agent who took trimmed piece, and NT be the agent (other than agent 1) who took the non-trimmed piece.
2. Agent NT divides cake 2 into three pieces they value equally.
3. the agents now choose in following order: Agent T, Agent 1 and then Agent NT

This method guarantees envy-freeness throughout. T selects from cake 2 before anyone else, so they cannot envy others. Agent 1 does not envy T's share because even if T took all the cake 2, the result would simply reconstruct the original piece agent 1 cut and valued as equal to their own allocation. Hence all agents finish without envy.

★ Optimization Problem

Now the allocation is expected to maximize total valuation.

$$\max_{A_1,A_2,A_3} \sum_{i=1}^{3} v_i(A_i)$$

constraints: (i) $A_i \cap A_j = \phi$, (ii) $\bigcup_{i=1}^{3} A_i = [0,1]$

(iii) $v_i(A_i) \geq 0 \ \forall i$ (iv) $v_i(A_i) \geq v_i(A_j) \ \forall i,j$.

★ Existence of solution.

→ Yes, a solution does exist. The selfridge Conway procedure guarantees at least one envy free allocation. So the feasible region F is non ~~empty~~ empty. The constraints on complete allocation and non-negativity confine the feasible region to a closed and bounded i.e compact subset of Euclidean space. The objective function is linear and hence continuous. The extreme value theorem then ensures that any continuous function on a non-empty compact set attains it's maximum, so an optimal solution is guaranteed.

Example:

Let $[0,1]$ be divided into three equal pieces $L = [0, \frac{1}{3}]$,

$M = [\frac{1}{3}, \frac{2}{3}]$, $R = [\frac{2}{3}, 1]$.

Define the valuation densities of each agent.

Agent	Density (L)	Density (M)	Density (R)
1	0.6	0.3	0.1
2	0.1	0.6	0.3
3	0.3	0.1	0.6

Example: $v_1(L) = 0.6$, $v_1(M) = 0.3$, $v_1(R) = 0.1$

Let's consider the allocation: ~~Agent 1 (S_1) L, Agent 2 (S_2) M, Agent 3 (S_3) R~~

$S_1 = L$; $S_2 = M$; $S_3 = R$

∴ Total valuation:

$$v_1(L) + v_2(M) + v_3(R) = 0.6 + 0.6 + 0.6 = 1.8.$$

Check envyness for agent 1:

$$v_1(L) = 0.6 \; ; \; v_1(M) = 0.3 \; ; \; v_1(R) = 0.1$$

~~So, $v_1(L) \geq v_2$~~ So, $v_1(S_1) \geq v_2(S_2) \geq v_1(S_3)$

By symmetry of the construction, the same idea holds for agent 2 and agent 3. Therefore the allocation is envy free.